

A Matrix-and-Probability Continuation of the Alaoglu–Erdős Program

Bulk Smith profiles, rational Loewner ensembles, singular-value hierarchies,
entropy saturation, and exact conditional proof certificates

Research continuation prepared from the supplied unified report

Literature and computation cutoff: 21 August 2026

Abstract

The Alaoglu–Erdős conjecture asks whether a real number x for which both 2^x and 3^x are integers must itself be an integer. The supplied unified report develops a large body of exact conditional structure, Lean-checked reductions, interpolation and determinant methods, a restricted Matrix Coefficient reformulation, Smith normalization, an HCIZ decomposition, signed-progression audits, and a first random-phase analysis. The present document is a separately written continuation and does not reproduce those developments.

The main new contribution is a matrix–probability framework that replaces a single determinant by the complete sequence of exterior powers. For an integer matrix A , the squared Hilbert–Schmidt norm of $\bigwedge^k A$ is exactly the total squared k -minor mass, and its normalization is the mean square of a uniformly sampled k -minor. Combining this identity with determinantal divisors, low-rank approximation, and diagonal gauge normalization yields a level-by-level comparison between the Smith profile of an integer evaluation matrix and the tail of the singular spectrum of its normalized analytic kernel. A multilevel modular-rank theorem then supplies explicit divisibility at every Smith level for arbitrary multivariate monomial evaluation matrices. An exact Smith normal form is derived for the geometric Vandermonde matrix (q^{ij}) , providing both a useful calibration and a warning that large Smith mass along a one-base ray is not by itself exceptional.

A second line uses the power Loewner kernel

$$K_\alpha(u, v) = \frac{u^\alpha - v^\alpha}{u - v}, \quad 0 < \alpha < 1.$$

Under a hypothetical counterexample, K_α is rational on two disjoint subsets of the dense group $\langle 2, 3 \rangle$. Its minors admit an exact Andreief–Cauchy integral representation as positive random-matrix partition functions. In the cluster limit they become Jacobi unitary ensemble integrals, evaluated by Selberg’s formula. The same kernel is a Gram kernel; orthogonal projection proves the universal hierarchy $\sigma_j = O(\varepsilon^{2(j-1)})$ for clustered cross matrices, and the Selberg asymptotic shows that these orders are sharp for fixed nondegenerate clusters. Combining this hierarchy with weighted Smith divisors gives a new family of exact conditional contradiction criteria at every compound-matrix level.

The probability-theoretic part proves an exact min-entropy anti-concentration bound for random prime-valuation vectors, then shows that a hypothetical counterexample saturates the associated entropy-rank bound: both integer outputs separately recover the two latent semigroup coordinates. Thus generic Littlewood–Offord reasoning can recover the known rank-two structure but has no entropy slack with which to force rank one. Two further no-go theorems show that natural positive-semidefinite/Hadamard-power constructions are infinitely divisible and therefore

blind to the arithmetic exponent, while the formal prime-log quadratic is sign-indefinite on the positive cone.

No unconditional proof is claimed. The report ends with four precise proof certificates, an obstacle ledger, and a computational program in which random minor sampling and random-prime scouting are used only to locate candidates that are then certified exactly.

Status

The conjecture was not closed. All statements labeled as theorems or propositions below are proved in this document from standard inputs explicitly cited. Numerical tables are checks, not proofs. Statements boxed as conditional targets are sufficient criteria whose unresolved hypotheses are stated visibly. The report’s strongest genuinely new synthesis is the *weighted Smith-profile versus singular-spectrum hierarchy*; it converts the determinant method from one scalar inequality into a complete family indexed by $1 \leq k \leq n$.

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1 Scope, inherited baseline, and non-duplication boundary

1.1 The problem and its four-exponentials boundary

The conjecture is

$$x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}. \quad (1)$$

A nonintegral solution is necessarily positive and irrational. Indeed, $x < 0$ would put 2^x strictly between 0 and 1, while a rational exponent $x = a/b$ in lowest terms for which 2^x is rational forces $b = 1$ by unique factorization. If such an x were algebraic irrational, Gelfond–Schneider would make 2^x transcendental, so every counterexample would be transcendental.

Writing the four algebraic exponentials as

$$\exp((\log 2) \cdot 1), \quad \exp((\log 2) \cdot x), \quad \exp((\log 3) \cdot 1), \quad \exp((\log 3) \cdot x),$$

shows why the problem sits exactly at the 2×2 four-exponentials boundary. The rows $\log 2, \log 3$ are $\mathbb{Q}$ -independent and, for nonintegral x , the columns $1, x$ are $\mathbb{Q}$ -independent. The Four Exponentials Conjecture would therefore settle (1), but the corresponding unconditional theorem currently needs a third row or column.

1.2 What the supplied report already contains

The source file audited for this continuation is

`/Article/alaoglu-erdos-unified-report/alaoglu-erdos-unified-report.tex`.

It has 16,614 lines and already contains, among much else:

- a kernel-checked zero-free region and exact conditional semigroup structure;
- canonical primitive odd cores, radical degrees, auxiliary-base spectra, and valuation lattices;
- generalized Vandermonde determinants, interpolation repulsion, and sparse power-curve rank;
- a restricted 2×2 Matrix Coefficient equivalence and formal prime-symbol determinant;
- factorial content, primitive maximal-minor Smith normalization, HCIZ accounting, and tropical assignment cascades;
- a signed-progression determinant with an exact denominator-tax identity;
- random-phase, Fejer-kernel, and microscopic pair-correlation certificates.

Those arguments are treated here as inherited infrastructure. Only the small amount of notation needed to state the new work is recalled.

1.3 Claim ledger

Label	Meaning in this report
Proved here	A complete paper proof is supplied. Most arguments use only Cauchy–Binet, Smith normal form, elementary module theory, the prime number theorem, Andreief’s identity, the Cauchy determinant, Selberg’s integral, and standard Hilbert-space approximation.
Inherited	The statement is used from the supplied unified report and is not reproved.
Classical input	A named theorem from the literature is used, with a bibliography entry.
Computational check	Exact symbolic or high-precision arithmetic confirms finite cases; the underlying theorem has a separate proof.
Conditional target	The displayed hypothesis would prove the conjecture, but the hypothesis is not established.
No-go	A tempting architecture is shown to be structurally incapable of detecting integrality without an additional arithmetic ingredient.

2 Literature audit through August 2026

The literature search emphasized primary sources and papers whose content touches the exact 2×2 boundary, matrix positivity, Smith profiles, or anti-concentration.

Work	Development	Relevance to the present program
Dasgupta–Kakde [7]	The Matrix Coefficient Conjecture, posted in 2024 and published in <i>Advances in Mathematics</i> 487 (2026), article 110753.	Gives a conceptual conjecture for singular matrices of logarithms of algebraic numbers. The supplied report proves that its restricted 2×2 instance is equivalent to (1). The present continuation instead studies integer evaluation matrices derived from a hypothetical singular logarithm matrix.
Massold [15]	New lower bounds and torus-intersection reductions for transcendence degrees of fields generated by arrays $\exp(\theta_i \kappa_j)$.	Places the surrounding problem in a modern transcendence-degree framework, but does not claim a resolution of the four-exponentials case. It therefore supplies context rather than a missing 2×2 estimate.
Bustos-Gajardo–Fokkink–Yassawi [5]	Version 3 (August 2026) proves a Gaussian-integer Cobham theorem without assuming Four Exponentials, under a structural alternative on the bases.	Demonstrates that some applications formerly routed through Four Exponentials can be bypassed by extra combinatorial structure. No analogous bypass is presently known for (1).

Work	Development	Relevance to the present program
Bhatia–Friedland–Jain [3]	Complete inertia analysis for power Loewner matrices; in particular positivity for $0 < \alpha < 1$.	Supplies the classical positivity backdrop for the rational Loewner construction below. Positivity alone is shown to be arithmetically blind, but the integral representation becomes useful when combined with denominator and Smith information.
Jain [12] and Khare [14]	Rank-two Hadamard powers and critical-exponent phenomena for positivity and total positivity; Khare’s current arXiv revision is version 2 (2021).	Motivates a search for matrix tests sensitive to nonintegral exponents. A no-go theorem below shows why the most natural 3-smooth Gram constructions are infinitely divisible and therefore cannot distinguish integer from noninteger powers.
Do–Nguyen–Phan–Tran–Vu [8]	A 2026 revision develops inverse anti-concentration on random permutations and nonsingularity applications.	Modern inverse Littlewood–Offord theory reinforces the principle that large concentration signals low-dimensional structure. The exact entropy analysis below shows that a hypothetical Alaoglu–Erdős semigroup already saturates that structural conclusion.
Wang–Stanley [21]	Local-to-global distributions of Smith normal forms of random integer matrices.	Provides a probabilistic comparison class. The evaluation matrices here are very far from random: residue-class collapse forces deterministic divisibility at every Smith level.

Audit note

The literature audit found no unconditional proof of the Four Exponentials Conjecture or of the Alaoglu–Erdős specialization as of 21 August 2026. The results below are therefore presented as new deductions and sharper proof targets, not as a claim to have silently crossed a known open boundary.

3 Counterexample normalization and the dense rational orbit

Assume for the remainder of the structural discussion that a nonintegral solution exists. Write

$$x = m + \alpha, \quad m = \lfloor x \rfloor, \quad 0 < \alpha < 1.$$

Set

$$r = 2^\alpha = \frac{2^x}{2^m} \in \mathbb{Q}_{>0}, \quad s = 3^\alpha = \frac{3^x}{3^m} \in \mathbb{Q}_{>0}. \quad (2)$$

The number α is irrational. Define the rank-two multiplicative group

$$\Gamma = \{2^a 3^b : a, b \in \mathbb{Z}\} \subset \mathbb{Q}_{>0}. \quad (3)$$

Because $\log 2 / \log 3 \notin \mathbb{Q}$, the additive group $\mathbb{Z} \log 2 + \mathbb{Z} \log 3$ is dense in $\mathbb{R}$, and hence Γ is dense in $\mathbb{R}_{>0}$. Moreover,

$$q^\alpha = r^a s^b \in \mathbb{Q}_{>0} \quad (q = 2^a 3^b \in \Gamma). \quad (4)$$

This elementary consequence of a counterexample drives the Loewner construction in section 10.

For integer evaluation matrices it is more convenient to retain the full exponent x . Put

$$M_2 = 2^x \in \mathbb{N}, \quad M_3 = 3^x \in \mathbb{N}.$$

Every pair $(n, k) \in \mathbb{N}^2$ gives the integer point

$$(X_{n,k}, Y_{n,k}) = (2^n M_2^k, 3^n M_3^k) = (2^{n+kx}, 3^{n+kx}). \quad (5)$$

For a monomial exponent $(a, b) \in \mathbb{N}^2$ its value is

$$X_{n,k}^a Y_{n,k}^b = \exp((a \log 2 + b \log 3)(n + kx)). \quad (6)$$

Thus every finite monomial evaluation matrix on the conditional semigroup is simultaneously an integer matrix and a sample of the strictly totally positive kernel $\exp(uv)$.

Part I

Exterior powers, Smith profiles, and random minors

4 Compound matrices as exact probability laws

Let $A \in \mathbb{R}^{n \times n}$ have singular values

$$\sigma_1(A) \geq \dots \geq \sigma_n(A) \geq 0.$$

For $1 \leq k \leq n$, write $\mathcal{I}_{n,k}$ for the k -subsets of $\{1, \dots, n\}$ and $A_{I,J}$ for the corresponding submatrix. Let

$$e_k(z_1, \dots, z_n) = \sum_{1 \leq i_1 < \dots < i_k \leq n} z_{i_1} \cdots z_{i_k}$$

be the k th elementary symmetric polynomial.

Theorem 4.1 (Exterior-power/minor identity). *For every real $n \times n$ matrix A ,*

$$\sum_{I, J \in \mathcal{I}_{n,k}} \det(A_{I,J})^2 = \left\| \bigwedge^k A \right\|_{\text{HS}}^2 = e_k(\sigma_1(A)^2, \dots, \sigma_n(A)^2). \quad (7)$$

Consequently, if I, J are independent uniform random k -subsets, then

$$\mathbb{E} \det(A_{I,J})^2 = \binom{n}{k}^{-2} e_k(\sigma_1(A)^2, \dots, \sigma_n(A)^2). \quad (8)$$

Proof. In the standard wedge bases, the matrix of $\bigwedge^k A$ is the k th compound matrix, whose (I, J) entry is $\det(A_{I,J})$. Its squared Hilbert–Schmidt norm is therefore the left side of (7). Singular-value decomposition commutes with exterior powers, and the singular values of $\bigwedge^k A$ are the products $\sigma_{i_1} \cdots \sigma_{i_k}$. Summing their squares gives the elementary symmetric polynomial. Dividing by the number $\binom{n}{k}^2$ of ordered pairs (I, J) proves (8). $\square$

The identity is not merely an analogy with probability: it makes the complete exterior spectrum an exact moment sequence of random minors. The determinant is only the endpoint $k = n$.

4.1 Determinantal divisors and arithmetic lower bounds

For a full-rank integer matrix $A \in \mathbb{Z}^{n \times n}$, define the k th determinantal divisor

$$\delta_k(A) = \gcd \{ |\det(A_{I,J})| : I, J \in \mathcal{I}_{n,k} \}. \quad (9)$$

If the Smith normal form has invariant factors

$$d_1 \mid d_2 \mid \cdots \mid d_n,$$

then

$$\delta_k(A) = d_1 d_2 \cdots d_k. \quad (10)$$

Let $N_k(A)$ be the number of nonzero k -minors.

Corollary 4.2 (Smith mass below exterior spectral mass). *For a full-rank integer matrix,*

$$e_k(\sigma(A)^2) \geq N_k(A) \delta_k(A)^2. \quad (11)$$

If every k -minor is nonzero, then

$$e_k(\sigma(A)^2) \geq \binom{n}{k}^2 \delta_k(A)^2, \quad \mathbb{E} \det(A_{I,J})^2 \geq \delta_k(A)^2. \quad (12)$$

Proof. Every nonzero k -minor is a nonzero multiple of $\delta_k(A)$ and therefore has squared magnitude at least $\delta_k(A)^2$. Sum and use theorem 4.1. $\square$

The semigroup matrices in (6), with distinct ordered row frequencies and node parameters, are strictly totally positive. Hence every minor is positive and the stronger form (12) applies simultaneously for all k .

5 Low-rank approximation versus the full Smith profile

The usual determinant method says that an integer determinant cannot be both nonzero and smaller than 1. The following theorem upgrades that statement to every exterior level.

Theorem 5.1 (Bulk Smith-profile obstruction). *Let $A \in \mathbb{Z}^{n \times n}$ be full rank and suppose every k -minor is nonzero. Assume*

$$A = B + R, \quad \text{rank } B \leq r < k, \quad \|R\|_2 \leq \varepsilon, \quad \|A\|_2 \leq S.$$

Then

$$\boxed{\sqrt{\binom{n}{k}} \delta_k(A) \leq S^r \varepsilon^{k-r}.} \quad (13)$$

Multiplying over $k = r + 1, \dots, n$ gives

$$\prod_{k=r+1}^n \left(\sqrt{\binom{n}{k}} \delta_k(A) \right) \leq S^{r(n-r)} \varepsilon^{(n-r)(n-r+1)/2}. \quad (14)$$

Proof. By Weyl's inequality, $\sigma_{r+1}(A), \dots, \sigma_n(A) \leq \varepsilon$. Put $\eta = \min(S, \varepsilon)$. Every product of k squared singular values contains at most r factors bounded by S^2 and at least $k - r$ further factors bounded by η^2 . Hence

$$e_k(\sigma(A)^2) \leq \binom{n}{k} S^{2r} \eta^{2(k-r)} \leq \binom{n}{k} S^{2r} \varepsilon^{2(k-r)}.$$

The lower bound (12) gives

$$\binom{n}{k}^2 \delta_k(A)^2 \leq e_k(\sigma(A)^2),$$

and combining the two inequalities proves (13). Summing the exponents of S and ε after multiplication gives (14). $\square$

Result proved in this continuation

A contradiction at any intermediate level $r < k < n$ is enough. This matters because the top determinant may lose arithmetic information through cancellation or gauge factors, while δ_k aggregates the divisibility of all k -minors. The product inequality (14) forces the entire initial Smith profile to fit inside one low-rank singular tail.

5.1 A general singular-envelope form

Uniformly bounding the whole tail by one number ε can be wasteful. The exact exterior identity immediately gives a sharper version.

Theorem 5.2 (Singular-envelope criterion). *Suppose $A \in \mathbb{Z}^{n \times n}$ has all k -minors nonzero, and suppose numbers $\tau_1 \geq \dots \geq \tau_n \geq 0$ satisfy $\sigma_j(A) \leq \tau_j$. Then*

$$\binom{n}{k} \delta_k(A) \leq \sqrt{e_k(\tau_1^2, \dots, \tau_n^2)} \leq \sqrt{\binom{n}{k}} \prod_{j=1}^k \tau_j. \quad (15)$$

Proof. The first inequality follows by placing the envelope into (12). For the second, every k -fold product of the nonincreasing τ_j^2 is at most $\prod_{j=1}^k \tau_j^2$, and there are $\binom{n}{k}$ terms. $\square$

The Loewner analysis in section 13 will produce the hierarchy $\tau_j \asymp \varepsilon^{2(j-1)}$, yielding the sharp compound exponent $k(k-1)$.

6 Diagonal gauges and weighted compound matrices

Integer evaluation matrices are often enormous because of row and column scales that are analytically irrelevant. Dividing them out destroys integrality, so a determinant-only argument pays the full denominator product. At intermediate exterior levels, the correct object is a weighted elementary symmetric sum.

Let

$$D = \text{diag}(\rho_1, \dots, \rho_n), \quad E = \text{diag}(\kappa_1, \dots, \kappa_n), \quad \rho_i, \kappa_j > 0,$$

and suppose

$$K = D^{-1}AE^{-1}. \tag{16}$$

For a subset I , write $\rho_I = \prod_{i \in I} \rho_i$, and similarly for κ_J .

Theorem 6.1 (Weighted exterior identity and Smith lower bound). *For every k ,*

$$e_k(\sigma(K)^2) = \sum_{I,J} \frac{\det(A_{I,J})^2}{\rho_I^2 \kappa_J^2}. \tag{17}$$

If A is an integer matrix and every k -minor is nonzero, then

$$e_k(\sigma(K)^2) \geq \delta_k(A)^2 e_k(\rho_1^{-2}, \dots, \rho_n^{-2}) e_k(\kappa_1^{-2}, \dots, \kappa_n^{-2}). \tag{18}$$

Proof. The k -minor of K indexed by I, J is $\det(A_{I,J})/(\rho_I \kappa_J)$. Apply theorem 4.1 and then bound every nonzero integer minor by $\delta_k(A)$ in absolute value. The remaining double sum factors into the two elementary symmetric polynomials. $\square$

Definition 6.2 (Gauge-invariant arithmetic mass). Define

$$\mathcal{A}_k(A; D, E) = \delta_k(A) \left(\frac{e_k(\rho^{-2}) e_k(\kappa^{-2})}{\binom{n}{k}} \right)^{1/2}. \tag{19}$$

Corollary 6.3 (Weighted Smith-tail criterion). *If $K = D^{-1}AE^{-1}$ has a rank- r approximation with error at most ε in operator norm and $\|K\|_2 \leq S$, then for $k > r$,*

$$\boxed{\mathcal{A}_k(A; D, E) \leq S^r \varepsilon^{k-r}}. \tag{20}$$

More generally, if $\sigma_j(K) \leq \tau_j$, then

$$\boxed{\mathcal{A}_k(A; D, E) \leq \prod_{j=1}^k \tau_j}. \tag{21}$$

Proof. Combine (18) with the upper bounds used in theorems 5.1 and 5.2. $\square$

At $k = n$, the quantity $\mathcal{A}_n$ is exactly $|\det K|$, because $\delta_n(A) = |\det A|$. For $k < n$, it records how much of the Smith mass survives after allowing the cheapest k row and column gauges. This is precisely the information erased by a determinant-only normalization.

6.1 Probabilistic interpretation of the weights

Choose random k -subsets with

$$\mathbb{P}(I) = \frac{\rho_I^{-2}}{e_k(\rho^{-2})}, \quad \mathbb{P}(J) = \frac{\kappa_J^{-2}}{e_k(\kappa^{-2})}.$$

Then

$$\mathbb{E} \det(A_{I,J})^2 = \frac{e_k(\sigma(K)^2)}{e_k(\rho^{-2})e_k(\kappa^{-2})} \geq \delta_k(A)^2. \quad (22)$$

Thus the diagonal gauge induces the exact importance-sampling distribution for minors. Random sampling can be used to locate promising levels and subsets, but the final inequality is deterministic and exact.

7 Multilevel modular rank forces bulk Smith divisibility

The supplied report obtains factorial divisors for coordinate lower sets. The next theorem is different: it applies to arbitrary monomial supports, controls every determinantal divisor, and uses all residue levels p^t .

Let $\mathcal{S} = \{\alpha_1, \dots, \alpha_n\} \subset \mathbb{N}^d$, and let $z_1, \dots, z_n \in \mathbb{Z}^d$. Define the square evaluation matrix

$$V_{ij} = z_j^{\alpha_i} = \prod_{\ell=1}^d z_{j,\ell}^{\alpha_{i,\ell}}. \quad (23)$$

Assume V has full rank over $\mathbb{Q}$.

For a prime p , let the p -adic valuations of the Smith factors be

$$0 \leq \nu_{p,1} \leq \dots \leq \nu_{p,n}.$$

Define

$$r_{p,t} = \# \{i : \nu_{p,i} < t\}. \quad (24)$$

Lemma 7.1 (Exact level decomposition). *For every k ,*

$$v_p(\delta_k(V)) = \sum_{t \geq 1} (k - r_{p,t})_+. \quad (25)$$

Proof. By (10), the left side is $\sum_{i=1}^k \nu_{p,i}$. Count the boxes in the Ferrers diagram of the nondecreasing sequence $\nu_{p,1}, \dots, \nu_{p,k}$ by horizontal levels. At level t , exactly $(k - r_{p,t})_+$ of the first k valuations are at least t . $\square$

Lemma 7.2 (Residue-class generator bound). *For the evaluation matrix (23),*

$$r_{p,t} \leq p^{td}. \quad (26)$$

Proof. Reduce the columns modulo p^t . A column depends only on the residue class of $z_j \in (\mathbb{Z}/p^t\mathbb{Z})^d$, so at most p^{td} distinct column vectors occur. Hence the image module of V over the local ring $R_t = \mathbb{Z}/p^t\mathbb{Z}$ is generated by at most p^{td} elements.

On the other hand, Smith reduction identifies that image with a direct sum of the nonzero cyclic modules $p^{\nu_{p,i}} R_t$ for $\nu_{p,i} < t$. By Nakayama's lemma, its minimal number of generators is exactly the number of such summands, namely $r_{p,t}$. Therefore $r_{p,t} \leq p^{td}$. $\square$

Theorem 7.3 (Multilevel modular-rank divisor). *For every $1 \leq k \leq n$,*

$$\delta_k(V) \text{ is divisible by } D_{k,d} := \prod_p p^{\sum_{t \geq 1} (k - p^{td})_+}. \quad (27)$$

Equivalently,

$$v_p(\delta_k(V)) \geq \sum_{t \geq 1} (k - p^{td})_+. \quad (28)$$

Proof. Insert (26) into (25). Only finitely many terms are nonzero. $\square$

Corollary 7.4 (Asymptotic bulk divisor). *For fixed $d \geq 1$,*

$$\log D_{k,d} = \left(\frac{d}{d+1} + o(1) \right) k^{1+1/d}. \quad (29)$$

If $r/n \rightarrow \rho \in [0, 1)$, then

$$\log \prod_{k=r+1}^n D_{k,d} = \left(\frac{d^2}{(d+1)(2d+1)} (1 - \rho^{2+1/d}) + o(1) \right) n^{2+1/d}. \quad (30)$$

For the bivariate case $d = 2$ the leading constants are $2/3$ and $\frac{4}{15}(1 - \rho^{5/2})$, respectively.

Proof. The $t = 1$ contribution is

$$\sum_{p^d < k} (k - p^d) \log p.$$

The prime number theorem and partial summation give

$$\sum_{p \leq y} \log p \sim y, \quad \sum_{p \leq y} p^d \log p \sim \frac{y^{d+1}}{d+1}.$$

Taking $y = k^{1/d}$ yields $\frac{d}{d+1} k^{1+1/d}$. The sum of the levels $t \geq 2$ is $O_d(k^{1+1/(2d)} \log k)$ and is lower order. Summing (29) over k and comparing with an integral gives (30). $\square$

Result proved in this continuation

For the conditional semigroup matrix (6), the columns are evaluations at integer points in two variables. Therefore every Smith level obeys

$$\delta_k \geq D_{k,2} = \prod_p p^{\sum_{t \geq 1} (k - p^{2t})_+}.$$

This lower bound is independent of the shape of the monomial support. It is weaker than specialized factorial formulas on some lower sets, but it is available simultaneously for all supports and all compound levels.

7.1 Finite calibration

The exact values of the universal lower bound approach their asymptotic constant slowly. The following values were computed directly from (27); they are included only as scale calibration.

k	$\log D_{k,2}$	$\log D_{k,2}/k^{3/2}$
30	58.843935	0.358113
100	490.516404	0.490516
300	2883.814682	0.554990
1000	19237.571186	0.608345
3000	104514.344239	0.636054
10000	645526.888780	0.645527

The limiting constant is $2/3$.

8 Exact Smith normal form of the geometric Vandermonde

The next exact computation is both a tool and a warning. Let

$$V_n(q) = (q^{ij})_{0 \leq i, j < n}, \quad q \in \mathbb{Z}, \quad q \geq 2. \quad (31)$$

Theorem 8.1 (Smith normal form of $V_n(q)$). *The Smith normal form of $V_n(q)$ is*

$$\text{diag}(s_0(q), s_1(q), \dots, s_{n-1}(q)), \quad (32)$$

where

$$s_i(q) = q^{\binom{i}{2}} \prod_{r=1}^i (q^r - 1), \quad s_0(q) = 1. \quad (33)$$

In particular $s_i(q) \mid s_{i+1}(q)$.

Proof. Replace the monomial row basis $1, X, X^2, \dots$ by the Newton basis

$$P_i(X) = \prod_{r=0}^{i-1} (X - q^r).$$

The change-of-basis matrix is lower unitriangular over $\mathbb{Z}$ and hence unimodular. Evaluation gives $P_i(q^j) = 0$ for $j < i$, while for $j \geq i$,

$$\begin{aligned} P_i(q^j) &= q^{\binom{i}{2}} \prod_{r=0}^{i-1} (q^{j-r} - 1) \\ &= s_i(q) \binom{j}{i}_q, \end{aligned}$$

where $\binom{j}{i}_q \in \mathbb{Z}$ is the Gaussian binomial coefficient. Thus the transformed matrix factors as DG , where $D = \text{diag}(s_i(q))$ and G is upper unitriangular over $\mathbb{Z}$. Right multiplication by $G^{-1} \in \text{GL}_n(\mathbb{Z})$ leaves D . Finally,

$$\frac{s_{i+1}(q)}{s_i(q)} = q^i (q^{i+1} - 1) \in \mathbb{Z},$$

so D is already in Smith order. □

Corollary 8.2 (Asymptotics and modular rank). *Let $c_q = \sum_{r \geq 1} \log(1 - q^{-r})$. Then*

$$\log s_i(q) = i^2 \log q + c_q + o(1), \quad \log \delta_k(V_n(q)) = \frac{\log q}{3} k^3 + O_q(k^2). \quad (34)$$

Moreover, for a prime p ,

$$\text{rank}_{\mathbb{F}_p} V_n(q) = \begin{cases} \min(n, 2), & p \mid q, \\ \min(n, \text{ord}_p(q)), & p \nmid q. \end{cases} \quad (35)$$

Proof. Expand $\log(q^r - 1) = r \log q + \log(1 - q^{-r})$ in (33) and sum. The modular-rank statement follows because the number of Smith factors not divisible by p is the rank modulo p . If $p \mid q$, only s_0, s_1 are prime to p . If $p \nmid q$, the first factor $q^r - 1$ divisible by p appears exactly when $r = \text{ord}_p(q)$. $\square$

Computational check 8.3 (Exact symbolic verification). Using SymPy's Smith normal form over $\mathbb{Z}$, formula (32) was verified for $q = 2, 3, 5$ and every $1 \leq n \leq 8$. For $n = 8$ the diagonal lists begin

$$\begin{aligned} q = 2 : & \quad 1, 1, 6, 168, 20160, 9999360, 20158709760, 163849992929280, \\ q = 3 : & \quad 1, 2, 48, 11232, 24261120, 475566474240, \\ & \quad 84129611558952960, 134068444202678083338240. \end{aligned}$$

The computation checks the implementation; the proof above is independent of it.

Obstruction 8.4 (One-base Smith mass is not a discriminator). The matrices $V_n(q)$ have enormous Smith profiles for every ordinary integer $q \geq 2$. Therefore a large determinantal divisor along a single geometric ray cannot distinguish a hypothetical exceptional integer $M_2 = 2^x$ from an unexceptional integer base. Any successful Smith argument must be genuinely mixed: it must retain interaction between the two bases, survive diagonal gauge normalization, or exploit modular rank defects not shared by generic integer rays.

9 Application to conditional semigroup evaluation matrices

Choose distinct monomial exponents $(a_i, b_i) \in \mathbb{N}^2$ and distinct semigroup indices $(n_j, k_j) \in \mathbb{N}^2$. Set

$$\lambda_i = a_i \log 2 + b_i \log 3, \quad (36)$$

$$t_j = n_j + k_j x, \quad (37)$$

$$A_{ij} = (2^{n_j} M_2^{k_j})^{a_i} (3^{n_j} M_3^{k_j})^{b_i} = e^{\lambda_i t_j}. \quad (38)$$

After sorting the λ_i and t_j , the kernel e^{uv} is strictly totally positive, so all minors of A are positive integers.

Choose centers λ_0, t_0 and write

$$u_i = \lambda_i - \lambda_0, \quad v_j = t_j - t_0.$$

Then

$$A_{ij} = e^{\lambda_0 t_0} e^{t_0 u_i} e^{\lambda_0 v_j} e^{u_i v_j}. \quad (39)$$

The first three factors are a common, row, and column gauge. The analytic core is

$$K_{ij} = e^{u_i v_j}. \quad (40)$$

Lemma 9.1 (Taylor low-rank approximation). *If $|u_i| \leq U$ and $|v_j| \leq V$, put $T = UV$. For every $r \geq 1$, the matrix*

$$B_r = \sum_{\ell=0}^{r-1} \frac{u^{\circ \ell} (v^{\circ \ell})^T}{\ell!} \quad (41)$$

has rank at most r and satisfies

$$\|K - B_r\|_2 \leq ne^T \frac{T^r}{r!}, \quad \|K\|_2 \leq ne^T. \quad (42)$$

Proof. Each summand in (41) has rank one. For $|z| \leq T$,

$$\left| e^z - \sum_{\ell < r} z^\ell / \ell! \right| \leq e^T \frac{T^r}{r!}.$$

The operator norm of an $n \times n$ matrix is at most n times its largest entry in absolute value. The same estimate with no subtraction gives $\|K\|_2 \leq ne^T$. $\square$

Distribute the common factor in (39) arbitrarily between the row and column gauges and let D, E be the resulting positive diagonal matrices with $A = DKE$. Combining theorems 6.3, 7.3 and 9.1 gives the following exact necessary condition for every hypothetical counterexample.

Criterion 9.2 (Bulk bounded-window certificate). *For every choice of distinct row exponents and semigroup nodes, every $r < k \leq n$ must satisfy*

$$D_{k,2} \left(\frac{e_k(D^{-2})e_k(E^{-2})}{\binom{n}{k}} \right)^{1/2} \leq n^k e^{kT} \frac{T^{r(k-r)}}{(r!)^{k-r}}. \quad (43)$$

Any finite configuration violating (43) proves the Alaoğlu–Erdős conjecture.

Proof. The modular theorem gives $\delta_k(A) \geq D_{k,2}$. Apply (20) with the bounds (42):

$$S^r \varepsilon^{k-r} \leq (ne^T)^r \left(ne^T \frac{T^r}{r!} \right)^{k-r},$$

which simplifies to the right side of (43). $\square$

Conditional proof target

The search problem is now finite and measurable: find two lattice sets whose logarithmic projections u_i and v_j lie in short windows, but for which the *weighted* Smith mass on the left of (43) remains large at some intermediate k . Uniform/global gauge factors are cancelled exactly, while nonuniform diagonal scales are explicitly penalized by $e_k(D^{-2})e_k(E^{-2})$. This is stricter than maximizing the full determinant and is well suited to modular-rank scouting.

Part II

Rational Loewner kernels and random-matrix partition functions

10 Rational cross-Loewner matrices under a counterexample

For $0 < \alpha < 1$, define the power divided-difference kernel

$$L_\alpha(u, v) = \begin{cases} \frac{u^\alpha - v^\alpha}{u - v}, & u \neq v, \\ \alpha u^{\alpha-1}, & u = v. \end{cases} \quad (44)$$

The classical Stieltjes representation is

$$L_\alpha(u, v) = c_\alpha \int_0^\infty \frac{t^\alpha}{(u+t)(v+t)} dt, \quad c_\alpha = \frac{\sin(\pi\alpha)}{\pi}. \quad (45)$$

It follows by subtracting the standard integral formula

$$u^\alpha = c_\alpha \int_0^\infty \frac{u}{u+t} t^{\alpha-1} dt.$$

If $u, v \in \Gamma$ are distinct, then (4) makes u^α, v^α rational, and therefore

$$L_\alpha(u, v) \in \mathbb{Q}. \quad (46)$$

The diagonal value contains the irrational factor α , so one must use two *disjoint* node sets rather than a principal Loewner matrix.

Let

$$0 < u_1 < \cdots < u_n, \quad 0 < v_1 < \cdots < v_n, \quad \{u_i\} \cap \{v_j\} = \emptyset,$$

with all nodes in Γ . Define the rational cross matrix

$$K_\alpha[\mathbf{u}; \mathbf{v}] = (L_\alpha(u_i, v_j))_{i,j=1}^n. \quad (47)$$

11 Andreief–Cauchy formula and strict total positivity

Theorem 11.1 (Positive n -fold integral for cross-Loewner minors). *For increasing positive node vectors $\mathbf{u}, \mathbf{v}$,*

$$\det K_\alpha[\mathbf{u}; \mathbf{v}] = \frac{c_\alpha^n}{n!} \Delta(\mathbf{u}) \Delta(\mathbf{v}) \quad (48)$$

$$\times \int_{(0,\infty)^n} \frac{\prod_{\ell=1}^n t_\ell^\alpha \Delta(\mathbf{t})^2}{\prod_{i,\ell} (u_i + t_\ell)(v_i + t_\ell)} d\mathbf{t}. \quad (49)$$

In particular, L_α is strictly totally positive on $(0, \infty) \times (0, \infty)$.

Proof. Apply Andreief's identity to (45) with $f_i(t) = (u_i + t)^{-1}$ and $g_j(t) = (v_j + t)^{-1}$:

$$\det \left(\int f_i(t) g_j(t) c_\alpha t^\alpha dt \right) = \frac{c_\alpha^n}{n!} \int \det(f_i(t_\ell)) \det(g_j(t_\ell)) \prod t_\ell^\alpha dt.$$

The Cauchy determinant formula gives

$$\det \left(\frac{1}{u_i + t_j} \right) = \frac{\Delta(\mathbf{u}) \Delta(\mathbf{t})}{\prod_{i,j} (u_i + t_j)},$$

and similarly for $\mathbf{v}$. Multiplying yields (49). The integral is finite and strictly positive, so every ordered minor is positive. $\square$

11.1 A random-matrix partition function

Set

$$Z_{n,\alpha} = \int_{(0,\infty)^n} \frac{\prod t_\ell^\alpha \Delta(\mathbf{t})^2}{\prod (1 + t_\ell)^{2n}} dt. \quad (50)$$

Normalize the integrand to a probability measure $\nu_{n,\alpha}$. Then (49) becomes

$$\det K_\alpha[\mathbf{u}; \mathbf{v}] = \frac{c_\alpha^n Z_{n,\alpha}}{n!} \Delta(\mathbf{u}) \Delta(\mathbf{v}) \quad (51)$$

$$\times \mathbb{E}_{\nu_{n,\alpha}} \prod_{\ell=1}^n \prod_{i=1}^n \frac{(1 + t_\ell)^2}{(u_i + t_\ell)(v_i + t_\ell)}. \quad (52)$$

The substitution $t = s/(1 - s)$ transforms $\nu_{n,\alpha}$ into the Jacobi unitary ensemble on $(0, 1)^n$ with weight

$$\prod_{\ell=1}^n s_\ell^\alpha (1 - s_\ell)^{-\alpha} \Delta(\mathbf{s})^2. \quad (53)$$

Both exponents α and $-\alpha$ are greater than -1 , so this is a genuine probability ensemble. Formula (52) is an exact positive expectation, with no determinant cancellation hidden inside it.

12 Selberg cluster asymptotics

Fix increasing real vectors $\boldsymbol{\xi}, \boldsymbol{\eta}$, with disjoint coordinate sets, and put

$$u_i = 1 + \varepsilon \xi_i, \quad v_j = 1 + \varepsilon \eta_j.$$

For sufficiently small positive ε , all nodes are positive and remain ordered.

Theorem 12.1 (Exact cluster exponent and Selberg constant). *As $\varepsilon \downarrow 0$,*

$$\det K_\alpha[\mathbf{u}; \mathbf{v}] = \kappa_{n,\alpha} \Delta(\boldsymbol{\xi}) \Delta(\boldsymbol{\eta}) \varepsilon^{n(n-1)} (1 + O(\varepsilon)), \quad (54)$$

where

$$\kappa_{n,\alpha} = \frac{c_\alpha^n}{n!} \prod_{j=0}^{n-1} \frac{\Gamma(j+1+\alpha) \Gamma(j+1-\alpha) \Gamma(j+2)}{\Gamma(n+j+1)}. \quad (55)$$

Proof. The two Vandermonde factors in (49) contribute

$$\Delta(\mathbf{u})\Delta(\mathbf{v}) = \varepsilon^{n(n-1)}\Delta(\boldsymbol{\xi})\Delta(\boldsymbol{\eta}).$$

Dominated convergence sends the integral to $Z_{n,\alpha}$. Under $t = s/(1-s)$, that integral is Selberg's integral with parameters

$$a = 1 + \alpha, \quad b = 1 - \alpha, \quad \gamma = 1.$$

Selberg's formula gives exactly the product in (55). The $O(\varepsilon)$ refinement follows by uniformly expanding the expectation in (52); for nodes in a fixed compact interval, the logarithm of every ratio factor is $O(\varepsilon)/(1+t)$. $\square$

Computational check 12.2 (High-precision check). For $\alpha = 0.37$, exponents $\xi_i = 2(i-1)$ and $\eta_j = 2j-1$, and $n = 2, 3, 4$, high-precision determinants were divided by the right side of (54). The ratios were:

n	$\varepsilon = 0.03$	$\varepsilon = 0.01$	$\varepsilon = 0.003$
2	0.94461349	0.98124702	0.99434327
3	0.86680514	0.95372266	0.98591325
4	0.76481854	0.91521995	0.97385274

The convergence is consistent with the proved asymptotic. It is not used as evidence for the conjecture.

12.1 Clusters inside the counterexample group

Because Γ is dense, there exist $q_\nu \in \Gamma$ with $q_\nu > 1$ and $q_\nu \rightarrow 1$. Fix disjoint increasing integer sets

$$a_1 < \cdots < a_n, \quad b_1 < \cdots < b_n.$$

Take

$$u_i = q_\nu^{a_i}, \quad v_j = q_\nu^{b_j}, \quad \varepsilon_\nu = \log q_\nu. \quad (56)$$

Then $u_i = 1 + a_i \varepsilon_\nu + O(\varepsilon_\nu^2)$ and similarly for v_j , so

$$\det K_\alpha[\mathbf{u}; \mathbf{v}] \sim \kappa_{n,\alpha} \Delta(\mathbf{a}) \Delta(\mathbf{b}) \varepsilon_\nu^{n(n-1)}. \quad (57)$$

Every entry is rational by (46). The problem is therefore reduced to comparing the explicit archimedean decay with the arithmetic cost of clearing denominators.

13 Gram projection and the singular-value hierarchy

The determinant asymptotic gives only the product of all singular values. The Gram structure gives each singular scale separately.

Let

$$\mathcal{H}_\alpha = L^2((0, \infty), c_\alpha t^\alpha dt), \quad \phi_u(t) = \frac{1}{u+t}. \quad (58)$$

Then

$$L_\alpha(u, v) = \langle \phi_u, \phi_v \rangle_{\mathcal{H}_\alpha}. \quad (59)$$

For $r \geq 1$, let P_r be the orthogonal projection onto

$$\mathcal{H}_r = \text{span} \left\{ (1+t)^{-1}, (1+t)^{-2}, \dots, (1+t)^{-r} \right\}, \quad (60)$$

and put $Q_r = I - P_r$.

Lemma 13.1 (Feature residual). *Fix $L > 0$. If $|u - 1| \leq L\varepsilon \leq 1/2$, then*

$$\|Q_r \phi_u\|_{\mathcal{H}_\alpha} \leq C_{r,\alpha,L} \varepsilon^r, \quad (61)$$

where one admissible constant is

$$C_{r,\alpha,L} = 2L^r (c_\alpha B(\alpha + 1, 2r + 1 - \alpha))^{1/2}. \quad (62)$$

Proof. The geometric expansion with exact remainder is

$$\frac{1}{1+t+(u-1)} = \sum_{m=0}^{r-1} \frac{(-(u-1))^m}{(1+t)^{m+1}} + \frac{(-(u-1))^r}{(1+t)^r(u+t)}.$$

The finite sum lies in $\mathcal{H}_r$. Since $u+t \geq (1+t)/2$,

$$\left| \frac{(-(u-1))^r}{(1+t)^r(u+t)} \right| \leq \frac{2(L\varepsilon)^r}{(1+t)^{r+1}}.$$

Squaring and integrating gives (62) through the beta integral. $\square$

Theorem 13.2 (Quadratic singular hierarchy for cross-Loewner clusters). *Let $K_\varepsilon = (L_\alpha(u_i, v_j))_{i,j=1}^n$, where all u_i, v_j lie within $L\varepsilon$ of 1. For every $0 \leq r < n$,*

$$\sigma_{r+1}(K_\varepsilon) \leq n C_{r,\alpha,L}^2 \varepsilon^{2r}, \quad (63)$$

with the convention that the $r = 0$ bound is replaced by any fixed uniform bound on $\|K\|_\varepsilon$. Hence there are constants T_j depending only on n, α, L such that

$$\boxed{\sigma_j(K_\varepsilon) \leq T_j \varepsilon^{2(j-1)} \quad (1 \leq j \leq n).} \quad (64)$$

Proof. Define the rank-at-most- r matrix

$$B_{ij} = \langle P_r \phi_{u_i}, P_r \phi_{v_j} \rangle.$$

Orthogonality gives

$$(K_\varepsilon - B)_{ij} = \langle Q_r \phi_{u_i}, Q_r \phi_{v_j} \rangle.$$

View the residual matrix as X^*Y , where X, Y are feature operators with n columns. The Frobenius norm of each is at most $\sqrt{n} C_{r,\alpha,L} \varepsilon^r$, so

$$\|K_\varepsilon - B\|_2 \leq n C_{r,\alpha,L}^2 \varepsilon^{2r}.$$

The Eckart–Young characterization of singular values gives (63). Set $r = j - 1$. $\square$

Corollary 13.3 (Sharp singular scales for nondegenerate affine clusters). *Fix increasing real vectors ξ, η with disjoint coordinate sets, and let*

$$K_\varepsilon = (L_\alpha(1 + \varepsilon\xi_i, 1 + \varepsilon\eta_j))_{i,j=1}^n.$$

For each $1 \leq j \leq n$ there are constants $c_j, C_j > 0$ and $\varepsilon_0 > 0$ such that

$$c_j \varepsilon^{2(j-1)} \leq \sigma_j(K_\varepsilon) \leq C_j \varepsilon^{2(j-1)} \quad (0 < \varepsilon < \varepsilon_0). \quad (65)$$

Proof. The upper bounds are theorem 13.2. Strict total positivity and theorem 12.1 give

$$\prod_{i=1}^n \sigma_i(K_\varepsilon) = \det K_\varepsilon \geq c \varepsilon^{n(n-1)}$$

for sufficiently small ε . For a fixed j , apply the upper bounds to every factor except σ_j :

$$\prod_{i \neq j} \sigma_i(K_\varepsilon) \leq C \varepsilon^{n(n-1)-2(j-1)}.$$

Dividing proves the lower bound in (65). □

Result proved in this continuation

The exponent is twice the naive Taylor exponent because the kernel error is an inner product of two orthogonal feature residuals. This is the matrix-theoretic reason the determinant has exponent

$$0 + 2 + 4 + \cdots + 2(n-1) = n(n-1).$$

For arbitrary bounded clusters the projection argument supplies the universal upper hierarchy; for fixed nondegenerate affine clusters, theorem 13.3 shows that every one of these orders is attained. Thus the Selberg determinant asymptotic and the Hilbert-space singular hierarchy match level by level.

14 Weighted Smith criteria for rational Loewner clusters

Let K_ε be a rational cross-Loewner matrix arising from nodes in Γ . Choose positive integer diagonal matrices

$$D = \text{diag}(d_1, \dots, d_n), \quad E = \text{diag}(e_1, \dots, e_n)$$

such that

$$A_\varepsilon = DK_\varepsilon E \in \mathbb{Z}^{n \times n}. \quad (66)$$

For example, one may clear denominators row by row, though an optimized row-column factorization can be much cheaper.

Criterion 14.1 (Loewner compound certificate). *Under a counterexample, every such clearing and every $1 \leq k \leq n$ must satisfy*

$$\delta_k(A_\varepsilon) \left(\frac{e_k(d_1^{-2}, \dots, d_n^{-2}) e_k(e_1^{-2}, \dots, e_n^{-2})}{\binom{n}{k}} \right)^{1/2} \leq C_{n,k,\alpha,L} \varepsilon^{k(k-1)}. \quad (67)$$

A violation for one finite rational cluster proves the Alaoglu–Erdős conjecture.

Proof. Apply the weighted singular-envelope inequality (21) and the bounds (64). The product of the first k envelopes has exponent $2 \sum_{j=1}^k (j-1) = k(k-1)$. $\square$

The endpoint $k = n$ is the ordinary denominator-versus-determinant comparison. The intermediate levels are new: they ask whether the rational clearing has enough primitive Smith mass to outrun the singular hierarchy before the full determinant is formed.

14.1 The optimized row-column denominator cost

For a rational matrix K , define

$$\mathfrak{D}_{\text{rc}}(K) = \min \left\{ \prod_i d_i \prod_j e_j : d_i, e_j \in \mathbb{N}_{>0}, \text{diag}(d_i) K \text{diag}(e_j) \in \mathbb{Z}^{n \times n} \right\}. \quad (68)$$

Strict total positivity makes the cleared determinant a positive integer, so

$$\det K \geq \mathfrak{D}_{\text{rc}}(K)^{-1}. \quad (69)$$

Combining with (57) gives an immediate sufficient condition.

Criterion 14.2 (Loewner denominator certificate). *Fix $n \geq 2$ and disjoint exponent sets $\mathbf{a}, \mathbf{b}$. If there exists a sequence $q_\nu \in \Gamma$, $q_\nu > 1$, $q_\nu \rightarrow 1$, such that the associated rational matrices satisfy*

$$\mathfrak{D}_{\text{rc}}(K_\nu) = o\left((\log q_\nu)^{-n(n-1)}\right), \quad (70)$$

then no counterexample exists.

Proof. The product of (69) with (57) would make the positive integer $\mathfrak{D}_{\text{rc}}(K_\nu) \det K_\nu$ tend to zero. $\square$

14.2 Why the obvious height estimate is far too expensive

Write a near-unity group element as

$$q = 2^u 3^{-v} > 1, \quad u, v \in \mathbb{N}.$$

Then $\log q = u \log 2 - v \log 3$. Continued-fraction convergents give sequences with $|\log q| = O(1/v)$, while the numerator and denominator of q have height $\exp(\Theta(v))$. Under a counterexample,

$$q^\alpha = r^u s^{-v},$$

whose naive height also grows exponentially in $u + v$. Clearing every entry of

$$\frac{q^{\alpha a_i} - q^{\alpha b_j}}{q^{a_i} - q^{b_j}}$$

by separate numerator and denominator heights therefore costs exponentially in $1/|\log q|$ along this natural sequence, whereas (70) permits only a fixed power of $1/|\log q|$.

Baker-type lower bounds for the nonzero linear form $u \log 2 - v \log 3$ show more generally that very small $|\log q|$ forces $u + v$ to grow at least as a positive power of $1/|\log q|$. Consequently, height-only denominator bounds are structurally too large. A successful Loewner proof must find *cancellation in the reduced row-column denominators* or additional Smith divisibility; it cannot merely clear the obvious powers of the numerators and denominators.

Audit note

This is an audit of the available bound, not a proof that the actual reduced denominators are always large. Unexpected cancellation is precisely what theorems 14.1 and 14.2 ask one to exploit.

Part III

Positivity tests and their structural blind spots

15 Hadamard powers: a rigorous infinite-divisibility no-go

The critical-exponent literature suggests testing whether a noninteger entrywise power loses positive semidefiniteness or total positivity. Integrality of the powered entries alone, however, does not force positivity. Worse, the most natural 3-smooth Gram matrices preserve positivity for *every* nonnegative real power.

Theorem 15.1 (Infinitely divisible 3-smooth Gram family). *Choose nonnegative integer vectors $a_1, \dots, a_n \in \mathbb{N}^r$ and $b_1, \dots, b_n \in \mathbb{N}^s$. Define*

$$H_{ij} = \langle a_i, a_j \rangle \log 2 + \langle b_i, b_j \rangle \log 3, \quad K_{ij} = e^{H_{ij}} = 2^{\langle a_i, a_j \rangle} 3^{\langle b_i, b_j \rangle}. \quad (71)$$

Then K is an integer matrix and

$$K^{\circ t} = (K_{ij}^t)_{ij} \succeq 0 \quad \text{for every } t \geq 0. \quad (72)$$

If x is a hypothetical counterexample, $K^{\circ x}$ is in addition an integer matrix.

Proof. The matrix H is positive semidefinite because it is a positive linear combination of two Gram matrices. By the Schur product theorem, every Hadamard power $H^{\circ m}$ is positive semidefinite. The entrywise exponential series converges in finite dimension and gives

$$K^{\circ t} = \exp^{\circ}(tH) = \sum_{m=0}^{\infty} \frac{t^m}{m!} H^{\circ m} \succeq 0.$$

If $M_2 = 2^x$ and $M_3 = 3^x$ are integers, then

$$(K^{\circ x})_{ij} = M_2^{\langle a_i, a_j \rangle} M_3^{\langle b_i, b_j \rangle} \in \mathbb{Z}.$$

□

The simplest example is $K = J + I$, whose diagonal entries are 2 and off-diagonal entries are 1. Then

$$K^{\circ x} = J + (2^x - 1)I \succeq 0$$

for every real $x \geq 0$. No critical-exponent transition is visible.

Obstruction 15.2 (Why positivity alone cannot prove integrality). The FitzGerald–Horn/Jain/Khare theory identifies matrices whose noninteger powers fail positivity or total positivity. But a negative determinant of an integer matrix is not a contradiction, and the natural matrices for which x -th powers are automatically Gram matrices lie in the infinitely divisible cone of theorem 15.1. A successful Hadamard-power argument therefore needs two ingredients simultaneously:

1. a 3-smooth integer test matrix outside the infinitely divisible cone; and
2. an independent arithmetic reason that its x -th Hadamard power must nevertheless be positive semidefinite or totally nonnegative.

Neither ingredient follows from $2^x, 3^x \in \mathbb{Z}$ alone.

16 The formal prime quadratic is not copositive

Let

$$M_2 = \prod_p p^{a_p}, \quad M_3 = \prod_p p^{b_p},$$

with finite nonnegative valuation vectors a, b . The supplied report introduces the formal homogeneous quadratic

$$\Phi(z) = z_2 \sum_p b_p z_p - z_3 \sum_p a_p z_p, \quad (73)$$

whose specialization $z_p = \log p$ is

$$\log 2 \log M_3 - \log 3 \log M_2.$$

A counterexample makes this specialization zero while Φ remains a nonzero polynomial.

Proposition 16.1 (Positive-cone sign change). *For a nonintegral candidate, M_2 has a prime divisor different from 2 and M_3 has a prime divisor different from 3. Consequently, Φ assumes both positive and negative values on the strict positive orthant and its zero cone intersects that orthant.*

Proof. If M_2 were a power of 2, then $2^x = M_2 = 2^m$ would force $x = m \in \mathbb{Z}$; similarly M_3 cannot be a pure power of 3. Fix all coordinates other than z_2, z_3 at positive values. Letting $z_3 \downarrow 0$ with z_2 fixed makes the first term in (73) dominate and gives a positive value, because b has support away from 3. Letting $z_2 \downarrow 0$ with z_3 fixed makes the second term dominate and gives a negative value, because a has support away from 2. A continuous path between the two points remains in the strict positive orthant, so it crosses the zero set. $\square$

The quadratic form matrix of Φ has rank at most four:

$$Q_\Phi = \frac{1}{2}(e_2 b^T + b e_2^T - e_3 a^T - a e_3^T). \quad (74)$$

Obstruction 16.2 (Inertia and copositivity cannot locate the prime-log point). The zero at $(\log p)_p$ is not forbidden by positive-cone geometry: every nonintegral candidate already forces other positive zeros of the same quadratic. Therefore an argument based only on the inertia of Q_Φ , copositivity, or separation of the positive orthant cannot prove the conjecture. One must use the arithmetic placement and rational independence properties of the specific point $(\log p)_p$.

Part IV

Probability, anti-concentration, and entropy saturation

17 Exact min-entropy anti-concentration for valuation vectors

Let P, Q be finite prime sets and define injective logarithmic maps

$$\Lambda_P(u) = \sum_{p \in P} u_p \log p, \quad \Lambda_Q(v) = \sum_{q \in Q} v_q \log q, \quad (75)$$

for integer vectors u, v . Injectivity is unique factorization.

For a discrete random variable Z , define its min-entropy by

$$H_\infty(Z) = -\log \max_z \mathbb{P}(Z = z). \quad (76)$$

Theorem 17.1 (Valuation anti-concentration). *Let U and V be independent random integer valuation vectors on P and Q . For nonzero real constants c, d ,*

$$\mathbb{P}(c\Lambda_P(U) = d\Lambda_Q(V)) \leq \min(\|\mathcal{L}(U)\|_\infty, \|\mathcal{L}(V)\|_\infty) = \exp(-\max(H_\infty(U), H_\infty(V))). \quad (77)$$

Proof. Condition on $V = v$. Since Λ_P is injective, the equation determines at most one value of U , so the conditional probability is at most the largest atom of U . Average over v . Interchanging U and V gives the other bound. $\square$

Corollary 17.2 (Uniform boxes). *If U and V are uniform on $\{0, \dots, H\}^r$ and $\{0, \dots, H\}^s$, then*

$$\mathbb{P}(c\Lambda_P(U) = d\Lambda_Q(V)) \leq (H + 1)^{-\max(r, s)}. \quad (78)$$

Hence the number of exact solution pairs is at most

$$(H + 1)^{\min(r, s)}. \quad (79)$$

Taking $c = \log 3$ and $d = \log 2$ makes the event exactly the common-exponent equation

$$\frac{\log m}{\log 2} = \frac{\log n}{\log 3}$$

for independently sampled integers supported on the selected prime sets.

17.1 Interpretation through inverse Littlewood–Offord theory

Modern anti-concentration theory says that unusually large point masses are usually caused by low-dimensional additive structure. Here the elementary injectivity argument is already sharp enough to expose the relevant dimension. If a family of exact pairs in two valuation boxes has order H^d , then (79) forces both effective valuation dimensions to be at least d . A hypothetical counterexample produces a two-parameter family and therefore naturally saturates the case $d = 2$. Anti-concentration recovers the known rank-two semigroup; by itself it does not collapse it to rank one.

18 Entropy growth equals valuation-lattice rank

The preceding point can be made exact in Shannon entropy.

Theorem 18.1 (Entropy-rank law for integer linear images). *Let $C \in \mathbb{Z}^{m \times r}$ have rank d over $\mathbb{Q}$, and let Z_H be uniform on $\{0, \dots, H\}^r$. Then*

$$H(CZ_H) = d \log(H + 1) + O_C(1). \quad (80)$$

The same leading term holds for min-entropy.

Proof. Choose d input coordinates and d output rows forming a nonsingular $d \times d$ submatrix. Once the remaining $r - d$ input coordinates are fixed, a specified output has at most one choice of the selected d coordinates. Thus every atom has probability at most $(H + 1)^{-d}$, giving

$$H_\infty(CZ_H) \geq d \log(H + 1).$$

For the reverse bound, choose d independent output rows. Their values lie in a box of side length $O_C(H)$, and they determine all remaining output coordinates over $\mathbb{Q}$. Hence the image has $O_C(H^d)$ points and $H(CZ_H) \leq d \log(H + 1) + O_C(1)$. Shannon entropy dominates min-entropy. $\square$

19 A hypothetical counterexample saturates all entropy bounds

Let x be a nonintegral solution and $M_2 = 2^x$, $M_3 = 3^x$. Let N, K be independent uniform random variables on $\{0, \dots, H\}$ and define

$$X_H = 2^N M_2^K, \quad Y_H = 3^N M_3^K. \quad (81)$$

Theorem 19.1 (Exact entropy saturation). *For every $H \geq 0$,*

$$H(X_H) = H(Y_H) = H(X_H, Y_H) = 2 \log(H + 1), \quad (82)$$

and therefore

$$I(X_H; Y_H) = 2 \log(H + 1). \quad (83)$$

Moreover, two valuation coordinates of either output already recover the latent pair (N, K) .

Proof. Because x is nonintegral, M_2 has an odd prime factor p . Then

$$v_p(X_H) = K v_p(M_2)$$

recovers K , and $v_2(X_H) = N + K v_2(M_2)$ then recovers N . Thus the map $(N, K) \mapsto X_H$ is injective. Similarly, a prime factor $q \neq 3$ of M_3 shows that $(N, K) \mapsto Y_H$ is injective. Each of $X_H, Y_H, (X_H, Y_H)$ is therefore an injective image of a uniform set of size $(H + 1)^2$, proving (82). The mutual information identity follows from $I(X; Y) = H(X) + H(Y) - H(X, Y)$. $\square$

Obstruction 19.2 (No Shannon-entropy slack). Both integer outputs separately contain all two-dimensional latent information, and they share it perfectly. Data processing, entropy subadditivity, and ordinary mutual-information inequalities are therefore equalities at leading order. Information theory can identify the rank-two semigroup but cannot force the rank-one integral branch without an additional inequality sensitive to the arithmetic values of the logarithms rather than merely to the valuation-lattice rank.

20 Generic random perturbations do not specialize

The formal polynomial Φ in (73) is nonzero for a hypothetical counterexample. If its variables are replaced by independent continuously distributed positive random variables, then

$$\mathbb{P}(\Phi(Z) = 0) = 0. \quad (84)$$

Likewise, random sign or random-phase models usually eliminate exact zeros. This is a correct generic statement but it says nothing about the named deterministic point $Z_p = \log p$. A low-degree polynomial can vanish at one prescribed point while being nonzero almost surely under every smooth perturbation.

This observation aligns with the random-phase analysis in the supplied report: almost-everywhere extinction is easy, while the arithmetically distinguished phase zero may support a rigid exceptional orbit. Probability becomes useful only when it produces a *quantitative deterministic inequality* that survives specialization, as the random-minor identity (8) does.

21 Randomized discovery, exact certification

Randomness can still accelerate the search for a finite contradiction without entering the proof.

21.1 Random-prime scouting

For a proposed integer evaluation matrix A , sample moderate primes p and compute

$$r_p = \text{rank}_{\mathbb{F}_p}(A \bmod p).$$

Every unexpectedly small r_p gives the exact divisor

$$v_p(\delta_k(A)) \geq (k - r_p)_+. \quad (85)$$

Computations modulo p^t or direct Smith routines can then certify the stronger multilevel profile. The random stage only identifies promising primes; the final rank calculation is exact.

21.2 Random minor sampling

For a normalized kernel $K = D^{-1}AE^{-1}$, sample k -subsets with the importance weights preceding (22). Large observed minors indicate a large exterior spectral mass and help identify which k carries the strongest arithmetic/analytic tension. Once a candidate is found, compute $\delta_k(A)$ and the relevant singular-value bound exactly or with certified intervals.

21.3 Search score

A convenient logarithmic score for a rank- r approximation is

$$\mathcal{S}_{k,r} = \log \mathcal{A}_k(A; D, E) - r \log S - (k - r) \log \varepsilon. \quad (86)$$

A rigorously positive score is a contradiction by (20). Floating-point values are only a filter; final candidates should use exact Smith divisors and directed-rounding norm bounds.

Part V

Synthesis and exact proof certificates

22 What the matrix and probability approaches genuinely add

The continuation yields five advances relative to the supplied report.

1. **The determinant becomes a profile.** The identities in sections 4 and 6 compare every determinantal divisor δ_k with the corresponding exterior singular mass. This exposes arithmetic information that can disappear at $k = n$.
2. **Modular collapse is multilevel.** The bound (27) uses all rings $\mathbb{Z}/p^t\mathbb{Z}$ and applies to arbitrary monomial supports. It provides a baseline Smith profile for every semigroup evaluation matrix.
3. **The Loewner determinant becomes a random-matrix ensemble.** Formula (52) is a positive expectation under a Jacobi unitary ensemble, and Selberg’s integral gives the exact cluster constant.
4. **The Loewner singular values are individually controlled.** Orthogonal projection in the Stieltjes feature space gives $\sigma_j = O(\varepsilon^{2(j-1)})$, matching the determinant exponent and enabling compound Smith tests.
5. **Probability’s limitation is quantified.** Min-entropy anti-concentration and the entropy-rank theorem show that a counterexample is not merely structured; it is entropy-extremal. Generic anti-concentration has no unused dimension to exploit.

23 Four exact sufficient certificates

23.1 Certificate A: weighted semigroup Smith-tail violation

Construct finite row and column lattice sets in (38). Compute a diagonal gauge and a rank- r approximation to the normalized exponential core. If, for any $k > r$,

$$\mathcal{A}_k(A; D, E) > S^r \varepsilon^{k-r},$$

then the conjecture follows. The universal lower bound $D_{k,2}$ may be substituted for $\delta_k(A)$, but a successful search will likely need extra modular rank defects.

23.2 Certificate B: Loewner denominator growth below the cluster exponent

Construct $q_\nu \in \Gamma$, $q_\nu \rightarrow 1$, and fixed disjoint exponent sets. If the optimized row-column denominator cost obeys (70), then the positive rational determinant is too small after clearing denominators.

23.3 Certificate C: an intermediate Loewner Smith-level violation

For one finite rational cluster, clear denominators to $A = DKE$ and prove that (67) fails at some $k < n$. This can succeed even when the full determinant inequality at $k = n$ is exactly balanced.

23.4 Certificate D: a genuinely mixed modular subgroup defect

For semigroup points modulo many primes, the generic residue-space bound is p^2 . If one can prove, for a positive-density or sufficiently weighted collection of primes and levels, that the actual column image is generated by substantially fewer than p^{2t} elements after removing row/column content, then (25) gives enhanced Smith mass. Combining that mass with a bounded-window singular tail through (20) is an exact proof route.

Conditional proof target

Certificate D appears the most compatible with the supplied report's valuation machinery. It asks for a modular shadow of the real rank-one relation that is strong enough at many primes, but it measures the shadow through a gauge-invariant Smith profile rather than through termwise tropical valuations.

24 Obstacle ledger

Approach	Exact gain	Remaining obstruction
Exterior/random minors	Converts all minors into the exact spectrum of $\bigwedge^k A$ and imports every determinantal divisor.	Need a configuration whose weighted Smith mass exceeds its analytic singular tail.
Universal modular divisor	Gives $\log \delta_k \geq (2/3 + o(1))k^{3/2}$ in two variables for arbitrary support.	On many natural square-grid scalings this is lower order than the gauge and analytic terms; extra rank defects are needed.
Geometric Vandermonde SNF	Exact closed form and cubic Smith growth.	The same phenomenon occurs for every ordinary integer base, so it carries no exceptional two-base information.
Loewner/Jacobi ensemble	Positive determinant, exact Selberg constant, and rational entries on counterexample orbits.	Naive denominator heights are exponentially larger than the polynomial cluster threshold.
Loewner singular hierarchy	Gives the optimal compound exponent $k(k-1)$.	Must show primitive Smith mass remains after optimized denominator clearing.
Hadamard critical exponents	Supplies many matrices whose noninteger powers change inertia or total positivity.	Powered entries being integers does not force positivity; natural Gram matrices are infinitely divisible.

Approach	Exact gain	Remaining obstruction
Formal quadratic matrix	Packages the logarithmic determinant into a rank-at-most-four form.	The form already has positive isotropic vectors, so copositivity cannot isolate the prime-log point.
Anti-concentration	Exact min-entropy bound and inverse-structure interpretation.	The conditional semigroup is a perfectly coupled rank-two extremizer, not an independent random family.
Shannon entropy	Gives an exact rank law for valuation images.	Both outputs separately recover all latent information, so every basic entropy inequality is saturated.

25 Prioritized computational and formalization program

25.1 Experiment 1: bulk Smith score on bounded logarithmic windows

Use lattice reduction to select n distinct pairs (a_i, b_i) for which $a_i \log 2 + b_i \log 3$ lies in a short interval, and similarly select (n_j, k_j) for $n_j + k_j x$ symbolically. Because x is unknown, the second selection should be expressed in terms of continued-fraction cells or the inherited least-generator parameter. For each combinatorial shape:

1. build the integer monomial evaluation matrix symbolically in M_2, M_3 ;
2. compute universal and support-specific modular rank bounds;
3. remove exact row/column gauges;
4. bound the singular tail of $e^{u_i v_j}$;
5. evaluate all scores (86), not only $k = n$.

25.2 Experiment 2: modular subgroup images

For many ordinary pairs (M_2, M_3) of comparable size, compute the image size of

$$(n, k) \mapsto (2^n M_2^k, 3^n M_3^k) \pmod{p^t}$$

for random primes and small t . The purpose is not to test the conjecture numerically, but to learn which algebraic features cause persistent image collapse after gauge normalization. Any conjectured deterministic rank bound can then be proved independently and inserted into (25).

25.3 Experiment 3: symbolic Loewner denominator factorization

Write $q = U/V$ and $q^\alpha = R/S$ in lowest terms and factor the cross entries

$$\frac{(R/S)^{a_i} - (R/S)^{b_j}}{(U/V)^{a_i} - (U/V)^{b_j}}$$

for fixed interlacing exponent sets. Optimize row/column denominator extraction exactly. The goal is to find systematic cyclotomic or resultant cancellation that makes $\mathfrak{D}_{\text{rc}}$ much smaller than entrywise height clearing predicts.

25.4 Experiment 4: formal verification

The following modules are natural Lean targets because their proofs are finite algebra or elementary analysis:

- exterior-power/minor identity and the weighted Smith-tail criterion;
- the level decomposition (25) and residue-generator bound;
- the exact q -Vandermonde Smith normal form;
- min-entropy anti-concentration and exact entropy recovery of (N, K) ;
- infinite divisibility of the 3-smooth Gram family;
- the positive-cone sign change of the formal quadratic.

The Andreief–Selberg and Hilbert-space projection sections are also formalizable but would require substantially more analytic infrastructure.

26 Conclusion

The matrix and probability viewpoints do not presently prove the Alaoglu–Erdős conjecture, but they sharpen the remaining wall in ways that the determinant endpoint alone cannot.

The central lesson is that arithmetic and analysis should be compared at every exterior level. An integer semigroup evaluation matrix has a full Smith profile forced by residue-class collapse. After the irrelevant row and column scales are removed, its analytic kernel has a rapidly decaying singular tail. The weighted compound inequalities state exactly how much arithmetic mass each tail can accommodate. A proof can therefore arrive from one intermediate minor size even when the full determinant is inconclusive.

The rational Loewner construction is the cleanest realization of this principle. Under a counterexample it produces strictly totally positive rational matrices on arbitrarily fine clusters. Their determinants are Jacobi-unitary-ensemble partition functions with an exact Selberg constant, and their singular values fall at the quadratic hierarchy $1, \varepsilon^2, \varepsilon^4, \dots$. What remains is an arithmetic theorem about optimized denominator/Smith content. The needed statement is no longer vague: it is precisely (67) or its endpoint (70).

Probability contributes an equally useful negative conclusion. Independent valuation vectors obey exact anti-concentration, but a hypothetical counterexample creates a perfectly coupled two-dimensional semigroup in which each output separately contains all latent entropy. Genericity arguments therefore cannot be specialized without a new quantitative bridge. Randomness is best used as a search device for modular rank defects and large minors, followed by exact certification.

Among the proposed next steps, the most promising is a hybrid: construct bounded logarithmic windows by lattice reduction, scout for systematic modular image collapse, and test the resulting

matrices with the complete weighted Smith profile. This directly joins the inherited valuation machinery to the new compound-matrix framework and avoids the one-base and common-content false positives isolated in this report.

A A compact proof of the prime-number-theorem asymptotic

For completeness, define

$$D_{k,d} = \prod_p p^{\sum_{t \geq 1} (k - p^{td})_+}.$$

Then

$$\log D_{k,d} = \sum_{t \geq 1} \sum_{p^{td} < k} (k - p^{td}) \log p.$$

At $t = 1$, with $y = k^{1/d}$,

$$\begin{aligned} \sum_{p \leq y} (k - p^d) \log p &= k \vartheta(y) - \sum_{p \leq y} p^d \log p \\ &= ky - \frac{y^{d+1}}{d+1} + o(k^{1+1/d}) \\ &= \frac{d}{d+1} k^{1+1/d} + o(k^{1+1/d}). \end{aligned}$$

For $t \geq 2$, the t th contribution is $O(k^{1+1/(dt)})$, and summing the geometrically decreasing exponents gives $O(k^{1+1/(2d)} \log k)$. This proves (29). The profile formula follows from

$$\sum_{k=r+1}^n k^{1+1/d} = \frac{d}{2d+1} (n^{2+1/d} - r^{2+1/d}) + o(n^{2+1/d}).$$

B Determinant constant from Selberg's integral

The cluster integral is

$$Z_{n,\alpha} = \int_{(0,\infty)^n} \frac{\prod t_i^\alpha \Delta(t)^2}{\prod (1+t_i)^{2n}} dt.$$

With $t_i = s_i/(1-s_i)$,

$$t_i^\alpha dt_i = (s_i^\alpha)(1-s_i)^{-\alpha-2} ds_i,$$

while the denominator contributes $(1-s_i)^{2n}$ and the squared Vandermonde contributes $(1-s_i)^{-2(n-1)}$. The net exponent is $-\alpha$. Hence

$$Z_{n,\alpha} = \int_{(0,1)^n} \prod s_i^\alpha (1-s_i)^{-\alpha} \Delta(s)^2 ds.$$

Selberg's formula with $a = 1 + \alpha$, $b = 1 - \alpha$, and $\gamma = 1$ yields

$$Z_{n,\alpha} = \prod_{j=0}^{n-1} \frac{\Gamma(j+1+\alpha)\Gamma(j+1-\alpha)\Gamma(j+2)}{\Gamma(n+j+1)},$$

which is (55) after multiplication by $c_\alpha^n/n!$.

C Reproducibility code for finite checks

The following script verifies the finite Smith examples, evaluates the Selberg cluster ratios, and computes the multilevel modular divisor table. It uses exact integer arithmetic for Smith forms and high precision only for the analytic calibration.

```
from __future__ import annotations
import math
import mpmath as mp
import sympy as sp
from sympy.matrices.normalforms import smith_normal_form
from sympy.polys.domains import ZZ

def predicted_snf(q: int, n: int) -> list[int]:
    out = []
    for i in range(n):
        s = q ** (i * (i - 1) // 2)
        for r in range(1, i + 1):
            s *= q**r - 1
        out.append(s)
    return out

def actual_snf(q: int, n: int) -> list[int]:
    V = sp.Matrix([[q ** (i * j) for j in range(n)]
                   for i in range(n)])
    D = smith_normal_form(V, domain=ZZ)
    return [abs(int(D[i, i])) for i in range(n)]

def loewner(alpha, x, y):
    if x == y:
        return alpha * x ** (alpha - 1)
    return (x**alpha - y**alpha) / (x - y)

def vandermonde(v):
    ans = mp.mpf(1)
    for i in range(len(v)):
        for j in range(i + 1, len(v)):
            ans *= v[j] - v[i]
    return ans

def selberg_constant(alpha, n):
    c = mp.sin(mp.pi * alpha) / mp.pi
    prod = mp.mpf(1)
    for j in range(n):
        prod *= (mp.gamma(j + 1 + alpha)
                 * mp.gamma(j + 1 - alpha)
                 * mp.gamma(j + 2)
                 / mp.gamma(n + j + 1))
    return c**n / mp.factorial(n) * prod

def modular_log_bound(k: int, d: int = 2) -> float:
    total = 0.0
```

```

    for p in sp.primerange(2, int(k ** (1 / d)) + 2):
        t = 1
        while p ** (t * d) < k:
            total += (k - p ** (t * d)) * math.log(p)
            t += 1
    return total

for q in (2, 3, 5):
    for n in range(1, 9):
        assert actual_snf(q, n) == predicted_snf(q, n)

mp.mp.dps = 80
alpha = mp.mpf("0.37")
for n in (2, 3, 4):
    a = list(range(0, 2 * n, 2))
    b = list(range(1, 2 * n + 1, 2))
    C = selberg_constant(alpha, n) * vandermonde(a) * vandermonde(b)
    for eps_text in ("0.03", "0.01", "0.003"):
        eps = mp.mpf(eps_text)
        X = [mp.e ** (eps * z) for z in a]
        Y = [mp.e ** (eps * z) for z in b]
        K = mp.matrix([[loewner(alpha, x, y) for y in Y] for x in X])
        ratio = mp.det(K) / (C * eps ** (n * (n - 1)))
        print(n, eps, ratio)

for k in (30, 100, 300, 1000, 3000, 10000):
    L = modular_log_bound(k, 2)
    print(k, L, L / k**1.5)

```

D Suggested theorem dependencies for formalization

A dependency-minimal order for formal verification is:

1. determinantal divisors and Smith-factor products;
2. compound matrices and Cauchy–Binet/Hilbert–Schmidt identities;
3. Weyl or min–max singular-value bounds for low-rank perturbations;
4. local-ring generator count for Smith factors modulo p^t ;
5. monomial-column congruence modulo p^t ;
6. Newton basis and Gaussian binomial integrality for theorem 8.1;
7. finite-distribution min-entropy and injective conditioning;
8. valuation recovery of semigroup coordinates;
9. Schur products and entrywise exponential limits;
10. elementary positive-orthant continuity for theorem 16.1.

This order isolates the core algebra from the analytic Loewner machinery and would already certify most of the new matrix/probability framework.

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